

Box Plot of Academic Performance Score

Figure Overview This box plot displays the distribution, central tendency, dispersion, and lower-tail outliers for the continuous variable `Academic_Performance_Score` across the analyzed student cohort.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median score centers around approximately 78.5, with the Interquartile Range (IQR) spanning from roughly 70.5 (Q_1) to 86.5 (Q_3), indicating high overall academic achievement across the middle 50% of students.
- **Negative Skewness & Outliers:** The distribution demonstrates a distinct left (negative) skew. While scores extend smoothly to an upper maximum of 100, the lower whisker terminates around 46, with a significant cluster of low-performing outliers stretching down to roughly 24.

Data Insights

- **Bimodal Risk Division:** Most students demonstrate strong academic competency, but the persistent tail of low-value outliers highlights a vulnerable subgroup experiencing severe academic underperformance.
- **Potential Correlation with Stress/Burnout:** Extreme low-end scores often correlate strongly with acute mental health strain, unmanaged burnout, or prolonged absenteeism, signaling a disconnect between general institutional outcomes and high-risk individuals.

Actionable Takeaways

- **Targeted Early-Intervention Thresholds:** Establish an automated risk trigger for students falling below the lower whisker boundary (~46) to provide immediate academic tutoring and mental health support.
- **Feature Engineering for Predictive Modeling:** Consider applying robust scaling or transformation techniques (e.g., Winsorization or log-based adjustments) to mitigate the influence of extreme low outliers during linear model training.

Box Plot of Age

Figure Overview This box plot depicts the overall demographic distribution, central tendencies, and spread of student ages within the dataset.

Key Findings & Statistical Trends

- **Symmetric Distribution:** The median age sits precisely at 19 years old, dividing the dataset into balanced upper and lower halves.
- **Broad Interquartile Range (IQR):** The middle 50% of the population spans between 16 (Q_1) and 22 (Q_3) years of age, reflecting an overall IQR of 6 years.
- **Full Range & Absence of Outliers:** The dataset ranges continuously from a minimum of 13 to a maximum of 25 years old, bounded cleanly by the whiskers.

without any statistical outliers beyond the $1.5 \times \text{IQR}$ limits.

Data Insights

- **Diverse Developmental Stages:** The broad age band indicates a dataset capturing both high school/secondary-level teenagers (13–17) and traditional university-aged adults (18–25).
- **Differential Stressors:** Environmental factors impacting academic performance and mental wellness likely vary significantly across age brackets (e.g., parental dependencies for younger students versus financial/career independence for older cohorts).

Actionable Takeaways

- **Age-Stratified Feature Engineering:** Segment the dataset into distinct age cohorts (e.g., Adolescents ≤ 17 vs. Young Adults ≥ 18) during modeling to evaluate whether stress predictors and failure risk factors interact differently across developmental stages.
- **No Outlier Truncation Needed:** The metric exhibits clean boundaries without extreme noise, requiring no data removal or capping before downstream regression or classification tasks.

Box Plot of Daily AI Tool Usage Hours

Figure Overview This box plot displays the statistical distribution, central tendency, dispersion, and upper-tail outliers for the continuous feature

`Daily_AI_Tool_Usage_Hours` across the surveyed student population.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median daily usage sits at approximately 2.5 hours, with the Interquartile Range (IQR) spanning from roughly 1.3 hours (Q_1) to 3.7 hours (Q_3).
- **Right Skewness & Heavy Tail:** The distribution exhibits a strong right (positive) skew. While the main upper whisker caps around 7.4 hours, a continuous stream of upper outliers extends beyond this boundary, reaching up to nearly 9.5 hours daily.

Data Insights

- **Bimodal Reliance Dynamics:** Most students maintain moderate interaction with AI tools (1–4 hours/day), integrating them as supplementary study aids. However, a distinct sub-cohort spends extreme duration (≥ 7.5 hours) relying on AI tools.
- **Over-reliance & Burnout Risk:** Unusually high daily usage hours may correlate with academic strain, time-management difficulties, or increased mental burnout, where students utilize AI to offset heavy course loads or deadline pressure.

Actionable Takeaways

- **Evaluate Non-Linear Relationships:** Analyze whether extreme usage hours (≥ 7.5) correlate with lower academic performance or higher mental distress scores using

non-linear modeling or binning techniques.

- **Cap or Transform Outliers for Linear Models:** Apply logarithmic transformation or capping at the 99th percentile to limit the leverage of heavy upper outliers in sensitive machine learning algorithms.

Box Plot of Daily Social Media Hours

Figure Overview This box plot evaluates the statistical distribution, central tendency, variance, and extreme values for the continuous variable `Daily_Social_Media_Hours` across the student cohort.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median daily social media usage sits at approximately 4.5 hours, with the Interquartile Range (IQR) spanning from around 2.8 hours (Q_1) to 6.2 hours (Q_3).
- **Right Skewness & Tail Distribution:** The feature exhibits a noticeable right (positive) skew. While the main upper whisker reaches up to roughly 11.2 hours, a distinct tail of outliers extends from 11.2 hours up to a maximum of 14 hours per day.

Data Insights

- **Substantial Daily Time Allocation:** Over half of the surveyed students spend more than 4.5 hours daily on social media platforms, representing a major allocation of non-academic discretionary time.
- **Extreme Usage Risks:** The cluster of extreme outliers (≥ 11.2 hours) points to a small subset of hyper-engaged users. In academic studies, such high levels of screen time often correlate with sleep deprivation, reduced study engagement, and heightened vulnerability to mental distress.

Actionable Takeaways

- **Cross-Feature Correlation:** Cross-examine `Daily_Social_Media_Hours` against `Academic_Performance_Score` and burnout metrics to identify threshold effects where social media consumption begins negatively impacting grades.
- **Feature Scaling & Transformation:** Apply robust scaling or log transformation to normalize the right-tail outliers before using this feature in linear regression models.

Box Plot of Mental Health Score

Figure Overview This box plot summarizes the central tendency, dispersion, and lower-tail outlier presence for the continuous variable `Mental_Health_Score` across the surveyed student population.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median score sits at approximately 73.5, with the Interquartile Range (IQR) spanning from around 67.0 (Q_1) to 79.5 (Q_3).

- **Negative Skewness & Outliers:** The distribution demonstrates a pronounced left (negative) skew. While upper scores reach smooth boundaries around 92.0, the lower whisker stops near 47.5, leaving a dense line of low-scoring outliers extending down to roughly 32.5.

Data Insights

- **Cohort Well-being Dichotomy:** Although the majority of students maintain high to moderate mental health scores (67–80+), a distinct tail of students faces acute mental wellness challenges.
- **Risk Marker for Student Distress:** The cluster of lower-tail outliers identifies individuals under severe psychological strain, making this variable a primary potential predictor for academic burnout and failure risk.

Actionable Takeaways

- **Institutional Support Triggers:** Flag students scoring below the lower whisker boundary (~47.5) as high-priority candidates for institutional wellness programs and academic counseling.
- **Model Outlier Management:** Utilize robust scale transformations or tree-based modeling techniques (e.g., Random Forests, XGBoost) that natively accommodate left-skewed features without losing predictive signals from lower-tail outliers.

Box Plot of Physical Activity Hours

Figure Overview This box plot evaluates the statistical distribution, central tendency, variance, and extreme values for weekly/daily `Physical_Activity_Hours` across the student cohort.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median physical activity time is approximately 1.1 hours, with the Interquartile Range (IQR) bounded between roughly 0.3 hours (Q_1) and 1.9 hours (Q_3).
- **Right Skewness & Tail Distribution:** The feature displays a distinct right (positive) skew. The upper whisker caps around 4.4 hours, with a continuous cluster of high-end outliers extending from 4.4 hours up to 5.0 hours.

Data Insights

- **Predominantly Sedentary Cohort:** The vast majority of students engage in relatively low amounts of physical activity, with 75% of the population logging under 2 hours.
- **Protective Factor Potential:** High physical activity levels often act as a buffer against stress and burnout; identifying students in the lower quantile (≤ 0.3 hours) may highlight groups more vulnerable to mental fatigue.

Actionable Takeaways

- **Bivariate Impact Analysis:** Test for interaction effects between `Physical_Activity_Hours` and `Mental_Health_Score` or `Academic_Performance_Score` to measure if higher physical activity mitigates academic stress.
- **Feature Transformation:** Apply log-transformation ($\log(x + 1)$) or robust scaling to smooth the right-skewed distribution prior to feeding this feature into linear regression models.

 Box Plot of Physical Health Score

Figure Overview This box plot summarizes the continuous distribution, central tendency, dispersion, and lower-tail outliers for `Physical_Health_Score` across the evaluated student population.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median physical health score sits at approximately 89.5, with the Interquartile Range (IQR) spanning from around 81.5 (Q_1) to 97.0 (Q_3).
- **Negative Skewness & Outlier Tail:** The distribution exhibits a strong left (negative) skew. While scores smoothly reach the upper limit of 100.0, the lower whisker stops near 58.0, leaving a persistent cluster of low-value outliers extending down to roughly 48.0.

Data Insights

- **High Baseline Physical Well-being:** The vast majority of the cohort maintains robust physical health scores above 80, suggesting generally strong overall physiological wellness in the population.
- **Vulnerable Subgroup Identification:** The lower-tail outliers (≤ 58.0) pinpoint a specific group experiencing significant physical health compromises, which could serve as a compound risk factor when paired with high academic stress or burnout.

Actionable Takeaways

- **Multivariate Risk Profiling:** Assess potential interactions between low physical health scores (≤ 58.0) and academic risk factors to determine if physical well-being mediates overall student burnout.
- **Robust Feature Scaling:** Use non-parametric scaling or tree-based algorithms (such as Gradient Boosting or Random Forests) to handle the left-skewed distribution without suffering degradation from low-end outliers.

 Box Plot of Sleep Hours

Figure Overview This box plot illustrates the continuous statistical distribution, central tendency, dispersion, and upper/lower-tail outliers for `Sleep_Hours` across the student population.

Key Findings & Statistical Trends

- **Symmetric Central Distribution:** The median sleep duration sits at approximately 6.6 hours per night, with the Interquartile Range (IQR) spanning symmetrically between roughly 5.7 hours (Q_1) and 7.4 hours (Q_3).
- **Bimodal Outlier Tails:** Whiskers extend cleanly to lower and upper boundaries of approximately 3.1 hours and 10.0 hours, respectively. Outliers exist on both tails, stretching down to 2.0 hours (severe sleep deprivation) and up to 11.2 hours (excessive hypersomnia).

Data Insights

- **Healthy Norm vs. Extremes:** While the central 50% of students fall within an acceptable rest range (5.7 to 7.4 hours), both extreme ends signal risk factors. Severe sleep deprivation (≤ 3.1 hours) is strongly linked to cognitive impairment and burnout, whereas excessive sleep (≥ 10.0 hours) often correlates with underlying mental health distress or lethargy.
- **Balanced Spread:** The symmetric nature of the box plot indicates a normal-like distribution around the median, making it well-suited for statistical modeling without heavy transformation.

Actionable Takeaways

- **Dual-Threshold Flagging:** Implement risk markers for students logging < 4.0 hours or > 10.0 hours to evaluate how extreme sleep durations compound mental health risks or academic failure rates.
- **Standard Scaling Application:** Given the symmetrical distribution, standard Z -score scaling or Min-Max scaling can be applied directly without needing log transformations for predictive models.

Box Plot of Academic Failure Risk

Figure Overview This box plot illustrates the statistical distribution, central tendency, and outlier prevalence for the binary classification metric `Academic_Failure_Risk` within the cohort dataset.

Key Findings & Statistical Trends

- **Zero Interquartile Range (IQR):** The lower quartile (Q_1), median (Q_2), and upper quartile (Q_3) all collapse directly onto 0.0, indicating that the vast majority of observations in the dataset are categorized as low/no risk (0).
- **Outlier Prevalence:** Values corresponding to 1.0 (high failure risk) appear strictly as isolated points beyond the main whisker boundary, highlighting severe class imbalance.

Data Insights

- **Class Imbalance Dynamics:** The distribution reveals that academic failure risk is a low-frequency occurrence across this population. While indicating strong general student performance, this extreme skewness presents a clear risk of majority-class bias for machine learning models.

- **Modeling Challenges:** Standard unweighted classification algorithms may struggle to learn decision boundaries for the minority class (1.0), potentially leading to high overall accuracy but poor recall for high-risk students.

Actionable Takeaways

- **Implement Resampling & Class Weighting:** Utilize techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or set cost-sensitive class weights during model training to prioritize sensitivity for the positive class.
- **Evaluate Precision-Recall Curves:** Rely on PR-AUC (Precision-Recall Area Under Curve) and F1-score rather than raw accuracy or standard ROC-AUC when evaluating baseline model performance.

Box Plot of Social Isolation Score

Figure Overview This box plot examines the continuous distribution, central tendency, dispersion, and extreme values for `Social_Isolation_Score` across the analyzed student cohort.

Key Findings & Statistical Trends

- **Central Tendency & IQR:** The median social isolation score centers at approximately 4.3, with the Interquartile Range (IQR) spanning from around 3.5 (Q_1) to 5.1 (Q_3).
- **Distribution & Outliers:** The distribution is near-symmetric around the median, with whiskers extending from approximately 1.1 to 7.6. A distinct cluster of high-end outliers appears on the upper tail, reaching up to roughly 8.4, alongside a minor low-end outlier around 1.0.

Data Insights

- **Moderate Overall Isolation:** The central 50% of students report moderate levels of social isolation (3.5 to 5.1), reflecting typical academic-life balance variation across the population.
- **High-Risk Vulnerability Group:** The upper-tail outliers (≥ 7.6) highlight a critical subgroup of highly isolated students. High social isolation is a well-documented accelerator for mental health decline, academic detachment, and heightened burnout risk.

Actionable Takeaways

- **Targeted Wellness Screening:** Establish behavioral intervention triggers for students recording a `Social_Isolation_Score` above 7.5 to provide proactive counseling and peer-support outreach.
- **Feature Engineering for Risk Prediction:** Evaluate interaction terms between high isolation scores (≥ 7.5) and screen time or academic performance metrics within predictive risk models.

Correlation Between Numerical Variables

Figure Overview This correlation heatmap visualizes the pairwise Pearson correlation coefficients across all numerical features in the dataset, identifying linear relationships between daily student habits, health metrics, academic performance, and failure risk.

Key Findings & Statistical Trends

- **Demographic Neutrality (Age):** Age shows virtually zero correlation ($r \approx -0.01$ to 0.02) with all other features, indicating that behavioral and health patterns are independent of student age in this cohort.
- **Strongest Positive Drivers:** Social_Isolation_Score correlates moderately with Daily_Social_Media_Hours ($r = 0.44$), while Physical_Activity_Hours positively impacts Physical_Health_Score ($r = 0.42$) and Mental_Health_Score ($r = 0.25$).
- **Primary Failure Risk & Performance Predictors:** Academic_Failure_Risk is most strongly driven by reduced Mental_Health_Score ($r = -0.36$), increased Daily_Social_Media_Hours ($r = 0.33$), and lower Academic_Performance_Score ($r = -0.30$). Concurrently, higher social media usage inversely impacts academic performance ($r = -0.39$) and mental health ($r = -0.40$).

Data Insights

- **Interconnected Vulnerability Loop:** Excessive social media consumption acts as a central negative anchor—strongly associating with social isolation, degraded mental/physical health, lower academic scores, and higher failure risk.
- **Protective Lifestyle Factors:** Adequate sleep ($r = 0.31$ with mental health, $r = 0.29$ with performance) and physical activity serve as key protective buffers against psychological distress and academic decline.

Actionable Takeaways

- **Multicollinearity & Feature Selection:** Since Age exhibits near-zero correlation across all targets, consider dropping it or treating it purely as a non-predictive demographic stratum to reduce feature space dimensions.
- **Prioritize High-Impact Predictors:** Focus feature engineering and model training on interaction terms combining Mental_Health_Score , Daily_Social_Media_Hours , and Social_Isolation_Score , as they hold the strongest directional coefficients for predicting academic failure risk.

 Count Distribution of Burnout Level

Figure Overview This bar chart illustrates the absolute frequency distribution across the ordinal categories of the categorical variable Burnout_Level (Low , Moderate , High , and Severe) within the student population.

Key Findings & Statistical Trends

- **Dominance of Low Burnout:** The Low burnout category forms the primary majority of the dataset, representing approximately 9,000 students.

- **Monotonic Decrease Across Severity:** Student frequency decays steadily as severity increases: `Moderate` burnout captures around 4,200 students, `High` burnout drops to roughly 1,350 students, and `Severe` burnout represents a small minority of approximately 450 students.

Data Insights

- **Severity Pyramid Dynamics:** The population follows a classic epidemiological pyramid structure, where mild/low symptoms dominate the baseline and critical distress cases remain localized to a smaller high-risk sub-population.
- **Target Stratification:** While the majority of students maintain manageable stress levels, the combined `High` and `Severe` groups (~1,800 students total) constitute the primary target cohort for urgent mental health intervention.

Actionable Takeaways

- **Ordinal Encoding Strategies:** Encode `Burnout_Level` using ordinal integer mapping (Low = 0, Moderate = 1, High = 2, Severe = 3) to preserve the natural progression of burnout intensity in machine learning models.
- **Class Weighting for Multiclass Tasks:** Apply inverse class weighting or sample re-balancing if predicting burnout levels directly to prevent models from underpredicting the critical `High` and `Severe` categories.

Count Distribution of Education Level

Figure Overview This count plot illustrates the categorical sample distribution of students across three distinct academic tiers: `College`, `High School`, and `University`.

Key Findings & Statistical Trends

- **Majority Stratum:** The `High School` segment represents the largest individual category in the dataset, accounting for approximately 5,700 students.
- **Balanced Higher-Education Split:** The remaining sample is evenly distributed between tertiary education levels, with `College` accounting for roughly 4,600 students and `University` comprising around 4,700 students.

Data Insights

- **Well-Balanced Representation:** The dataset exhibits a balanced representation across all three education levels without extreme class imbalance, ensuring sufficient statistical power within each subgroup.
- **Demographic Structural Alignment:** The distribution aligns with expected academic progression pathways, offering an ideal setup to cross-evaluate performance metrics and burnout indicators across varying institutional environments.

Actionable Takeaways

- **One-Hot / Categorical Encoding:** Apply One-Hot Encoding to `Education_Level` prior to feature selection or model training to prevent non-ordinal bias.
- **Subgroup Comparative Analysis:** Conduct stratified evaluations (e.g., grouping by `Education_Level`) to test if predictors of burnout and academic failure risk function consistently between high school and tertiary students.

Count Distribution of Gender

Figure Overview This count plot displays the categorical frequency distribution across the gender identities recorded within the student cohort dataset.

Key Findings & Statistical Trends

- **Binary Parity:** The dataset exhibits near-perfect parity between `Male` and `Female` categories, with `Male` representing approximately 7,280 students and `Female` accounting for roughly 7,150 students.
- **Minority Class Representation:** The `Non-binary` category represents a small minority within the cohort, capturing approximately 600 students (roughly 4% of the total sample).

Data Insights

- **Balanced Binary Demographic:** The nearly 1:1 ratio between male and female students minimizes demographic bias across the primary classes, enabling unbiased cross-gender comparisons of performance and mental health metrics.
- **Underrepresented Group Considerations:** While the inclusion of non-binary identities reflects comprehensive data collection, its low sample size requires careful handling to prevent overfitting or unreliable statistical inferences for this specific group.

Actionable Takeaways

- **Categorical Encoding:** Implement One-Hot Encoding for the `Gender` feature during preprocessing to prevent imposing false ordinal hierarchy in downstream ML models.
- **Stratified Validation:** Ensure cross-validation schemes use stratified sampling to maintain consistent proportions of the `Non-binary` class across train/test splits.

Distribution of Academic Failure Risk

Figure Overview This histogram and Kernel Density Estimation (KDE) plot depict the continuous density distribution and frequency counts for the target feature `Academic_Failure_Risk` across the surveyed student population.

Key Findings & Statistical Trends

- **Extreme Zero-Inflation:** The vast majority of observations aggregate at 0.0, capturing approximately 14,100 students who are not classified as at-risk for academic failure.

- **Minority Positive Class Density:** The positive risk outcome (1.0) accounts for a small peak representing approximately 900 students, confirming a significant class imbalance within the dataset (roughly 15:1 ratio).

Data Insights

- **Target Class Imbalance Dynamics:** The severe skew toward 0.0 highlights that failure risk is a low-probability event across the broader cohort, reflecting healthy overall student retention but posing classification challenges.
- **KDE Smoothing Artifact:** The continuous KDE curve artificially spreads density between 0.0 and 0.2, which is an artifact of density estimation on discrete binary data (0/1) rather than a reflection of continuous underlying risk values.

Actionable Takeaways

- **Metric Selection for Evaluation:** Avoid standard classification accuracy as a primary metric; utilize Precision-Recall AUC (PR-AUC) or F1-Score to effectively monitor positive-class detection performance.
- **Apply Imbalance Mitigation Strategies:** Employ class-weighted loss functions (e.g., `scale_pos_weight` in XGBoost/LightGBM) or oversampling algorithms (such as SMOTE) during model execution to prevent majority-class bias.

Distribution of Academic Performance Score

Figure Overview This histogram overlaid with a Kernel Density Estimation (KDE) curve illustrates the frequency distribution, skewness, and density profile of `Academic_Performance_Score` across the student population.

Key Findings & Statistical Trends

- **Primary Peak & Central Tendency:** The distribution peaks strongly between scores of 75 and 83, matching the cohort's average performance level and exhibiting a broad, high-density cluster around 650 students per bin.
- **Negative Skewness & Ceiling Peak:** The metric displays a clear left (negative) skew, with counts tapering gradually down toward the minimum score of ~24. Notably, a secondary spike occurs near the upper boundary of 100, reflecting a concentration of high-achieving students.

Data Insights

- **High Baseline Competency:** Most students demonstrate solid academic achievement, maintaining scores concentrated above 70.
- **Subgroup Polarization:** The left-hand tail captures a minority group struggling with sub-50 scores, while the right-hand ceiling peak indicates a distinct segment of top performers reaching maximum score limits.

Actionable Takeaways

- **Targeted Academic Remediation:** Establish automated academic support triggers for students scoring below 60 to prevent further drop toward the severe low-end

tail.

- **Non-Linear Model Alignment:** Utilize tree-based regression models (e.g., Random Forest, XGBoost) or apply polynomial feature transformations to handle the slight left skew and ceiling effect at 100.

Distribution of Age

Figure Overview This histogram and Kernel Density Estimation (KDE) plot examine the sample distribution, discrete counts, and overall dispersion of student ages within the dataset.

Key Findings & Statistical Trends

- **Uniform Discrete Distribution:** The `Age` variable follows a near-perfect uniform distribution across discrete integer values ranging from 13 to 25 years old.
- **Balanced Class Proportions:** Each individual age bin contains a consistently stable representation, averaging between approximately 1,100 and 1,240 students per age year, with a flat KDE trend line across the entire spectrum.

Data Insights

- **Unbiased Demographic Sampling:** The synthetic or controlled nature of the sample guarantees equal representation across secondary school (ages 13–17) and post-secondary/university cohorts (ages 18–25).
- **Minimal Direct Predictive Power:** Because `Age` is uniformly distributed and unskewed, it functions as a neutral baseline feature rather than a standalone indicator for localized target variance.

Actionable Takeaways

- **Cohort-Based Grouping:** Re-encode the continuous/discrete age values into functional developmental bins (e.g., `Adolescent` : 13–17, `Undergraduate` : 18–21, `Postgraduate/Adult` : 22–25) to evaluate cohort-level interaction effects.
- **Baseline Conditioning:** Use `Age` for stratified splitting or controlling demographic bias, ensuring that predictive modeling models do not over-index on specific age segments.

Distribution of Daily AI Tool Usage Hours

Figure Overview This histogram and Kernel Density Estimation (KDE) plot examine the frequency distribution, central tendency, right-tail skew, and zero-inflation spike for continuous `Daily_AI_Tool_Usage_Hours` across the student population.

Key Findings & Statistical Trends

- **Zero-Inflation Spike:** A distinct spike occurs at exactly 0.0 hours, representing a non-user cohort of approximately 1,500 students who do not integrate AI tools into their daily routine.
- **Unimodal Secondary Peak & Right Skew:** For active users, the distribution forms a smooth, broad peak centered between 2.0 and 3.0 hours daily, tapering off gradually

in a positive right-skew up to a maximum near 8.0 to 9.0 hours.

Data Insights

- **Adoption Dichotomy:** The feature displays a classic zero-inflated continuous pattern, clearly separating non-adoptive students (0.0 hours) from regular users whose engagement typically spans 1.5 to 4.0 hours per day.
- **Heavy Usage Minority:** Only a small fraction of the student body logs extreme usage (≥ 6.0 hours), suggesting that heavy AI dependence remains localized to a minor subset of the sample.

Actionable Takeaways

- **Two-Step Feature Engineering:** Create an explicit binary indicator (`Uses_AI_Tools` : 0 vs. 1) alongside the continuous hour metric to capture structural differences between non-users and active users in downstream predictive models.
- **Non-Linear Transformations:** Apply a log transformation ($\log(x + 1)$) or zero-inflated model architectures (such as hurdle models or decision trees) to properly handle the zero mass and positive right skew.

 Distribution of Daily Social Media Hours

Figure Overview This histogram and Kernel Density Estimation (KDE) plot detail the frequency distribution, central tendency, right-tail skewness, and zero-value clustering for `Daily_Social_Media_Hours` across the student population.

Key Findings & Statistical Trends

- **Primary Peak & Central Concentration:** The continuous distribution peaks strongly between 4.0 and 5.0 hours daily, reaching nearly 680 students per bin, representing the dominant usage habit across the cohort.
- **Zero-Spike & Right-Tail Skew:** A localized zero-inflation spike appears at 0.0 hours (~680 students), representing non-users. Beyond the central peak, the distribution shows a long right-tail skew extending up to 14.0 hours per day.

Data Insights

- **High Average Engagement:** The majority of students spend between 3.0 and 7.0 hours daily on social media, making it a major component of their non-academic routine.
- **Extreme Usage Outliers:** A small subgroup logs extreme usage above 9.0 hours daily, which often correlates with sleep disruption, social isolation, and academic risk.

Actionable Takeaways

- **Dual-Feature Preprocessing:** Create a binary flag (`Is_Social_Media_User` : 0 vs. 1) to account for the zero spike alongside the continuous hour feature.
- **Log Transformation:** Apply a $\log(x + 1)$ transformation to compress the long right-tail skew before using this variable in linear modeling algorithms.

Distribution of Mental Health Score

Figure Overview This histogram overlaid with a Kernel Density Estimation (KDE) curve illustrates the continuous frequency distribution, central tendency, and left-tail skewness for `Mental_Health_Score` across the student population.

Key Findings & Statistical Trends

- **Primary Peak & Central Tendency:** The distribution forms a broad, high-density peak centered between scores of 74 and 79, reaching nearly 680 students per bin.
- **Negative Skewness & Lower-Tail Extension:** The data exhibits a clear left (negative) skew. While scores drop off steeply above 85 up to a maximum near 92, the lower tail extends gradually down to around 32.

Data Insights

- **Cohort Mental Well-Being Baseline:** The majority of surveyed students maintain moderate-to-high mental health scores, concentrated predominantly between 67 and 80.
- **Distressed Subgroup Identification:** The long lower tail highlights a vulnerable subgroup of students scoring below 50, reflecting heightened psychological distress and potential academic burnout risk.

Actionable Takeaways

- **Intervention Thresholds:** Establish proactive support triggers for students recording a `Mental_Health_Score` below 50 to connect them with institutional wellness resources.
- **Feature Engineering & Transformation:** Apply robust scaling or non-linear algorithms (such as tree-based models) to handle the left-skewed tail without losing critical predictive signals from low-scoring outliers.

Distribution of Physical Activity Hours

Figure Overview This histogram overlaid with a Kernel Density Estimation (KDE) curve illustrates the continuous frequency distribution, zero-inflation spike, and right-tail decay for `Physical_Activity_Hours` across the student population.

Key Findings & Statistical Trends

- **Zero-Inflation Spike:** A dominant spike appears at exactly 0.0 hours, capturing over 3,100 students who report zero daily physical activity.
- **Unimodal Secondary Spread & Right Skew:** For active individuals, the distribution forms a broad peak centered between 0.8 and 1.5 hours daily before tapering off in a right-skewed tail up to a maximum of 5.0 hours.

Data Insights

- **High Sedentary Proportion:** Over 20% of the surveyed population logs no daily physical activity, presenting a widespread sedentary lifestyle baseline across the cohort.
- **Low Average Engagement:** Even among active students, daily activity remains relatively modest, with very few logging more than 3.0 hours.

Actionable Takeaways

- **Binary Feature Engineering:** Create a dedicated indicator variable (`Is_Physically_Active` : 0 vs. 1) to model structural differences between non-active and active students.
- **Distribution Scaling:** Apply a $\log(x + 1)$ transformation or utilize tree-based algorithms to effectively manage the heavy zero mass and positive skew.

 Distribution of Sleep Hours

Figure Overview

This histogram with a Kernel Density Estimation (KDE) curve shows the distribution of `Sleep_Hours` among students, including the central tendency, spread, and overall shape of the data.

Key Findings & Statistical Trends

- **Central Distribution:** Most students sleep between approximately **5 and 8.5 hours per day**, with the distribution centered around **6.5–6.8 hours**.
- **Approximately Symmetric Distribution:** The data follows a roughly **bell-shaped and balanced distribution**, with no strong skew toward either low or high sleep durations.
- **Range:** Sleep duration ranges from approximately **2 to 11 hours per day**.
- **Extreme Values:** A small number of students report very low sleep duration (**below 4 hours**) or unusually high sleep duration (**above 9.5 hours**).

Data Insights

- **Typical Sleep Pattern:** Most students have sleep durations around **6–8 hours**, representing the typical range within the dataset.
- **Sleep-Deprived Group:** Students sleeping less than **4–4.5 hours** form a smaller subgroup that may be useful to investigate when studying academic performance and mental health.
- **Balanced Distribution:** Since the variable is approximately symmetric, there is no obvious need for a logarithmic transformation.

Actionable Takeaways

- **Sleep Monitoring:** Consider examining students with less than **4.5 hours of sleep** as a potentially higher-risk group when analyzing academic and health outcomes.

- **Machine Learning:** `Sleep_Hours` can generally be used directly in many models. For algorithms that benefit from standardized numerical features, `StandardScaler` can be applied.
- **Further Analysis:** Compare `Sleep_Hours` with `Mental_Health_Score`, `Academic_Performance_Score`, and `Academic_Failure_Risk` to investigate possible relationships.

 Distribution of Social Isolation Score

Figure Overview This histogram overlaid with a Kernel Density Estimation (KDE) curve presents the continuous frequency distribution, central tendency, and symmetric spread for `Social_Isolation_Score` across the surveyed student population.

Key Findings & Statistical Trends

- **Symmetric Gaussian Distribution:** The metric exhibits a classic bell-shaped (normal) profile centered around a mean of approximately 4.2, peaking at nearly 680 students per bin.
- **Balanced Spread Across Range:** The scores span symmetrically from a lower bound near 1.0 to an upper tail maximum around 8.0, with the vast majority of students scoring between 2.5 and 6.0.

Data Insights

- **Moderate Overall Baseline:** Most students experience moderate levels of social isolation, clustering safely away from the extreme high end of the scale.
- **Symmetric Extremes:** Students feeling virtually no isolation (≤ 2.0) and those experiencing severe isolation (≥ 6.5) form equal, symmetric tail segments, making isolation risk an even-spectrum factor across the cohort.

Actionable Takeaways

- **Standard Preprocessing:** Apply standard normalization (`StandardScaler` / Z-score transformation) directly, as the feature adheres to a normal distribution without requiring non-linear transforms.
- **High-Isolation Intervention Threshold:** Set an automated flag for students recording a `Social_Isolation_Score` above 6.0 to trigger wellness check-ins and community integration programs.

 Distribution of Physical Activity Hours

Figure Overview This histogram overlaid with a Kernel Density Estimation (KDE) curve illustrates the continuous frequency distribution, zero-inflation spike, and right-tail decay for `Physical_Activity_Hours` across the student cohort.

Key Findings & Statistical Trends

- **Zero-Inflation Spike:** A dominant spike appears at exactly 0.0 hours, capturing over 3,100 students who report zero daily physical activity.

- **Unimodal Secondary Spread & Right Skew:** For active individuals, the distribution forms a broad peak centered between 0.8 and 1.5 hours daily before tapering off in a right-skewed tail up to a maximum of 5.0 hours.

Data Insights

- **High Sedentary Proportion:** Over 20% of the surveyed population logs no daily physical activity, presenting a widespread sedentary lifestyle baseline across the cohort.
- **Low Average Engagement:** Even among active students, daily activity remains relatively modest, with very few logging more than 3.0 hours.

Actionable Takeaways

- **Binary Feature Engineering:** Create a dedicated indicator variable (`Is_Physically_Active` : 0 vs. 1) to model structural differences between non-active and active students.
- **Distribution Scaling:** Apply a $\log(x + 1)$ transformation or utilize tree-based algorithms to effectively manage the heavy zero mass and positive skew.